

Math camp exam*

Solutions

2022

Department of Economics, Columbia University

This is a timed remote exam. You have 3 hours to submit your answers via email to the TA (and cc the instructor). Total points are 100. Please justify your answers. Partial credit will be awarded in case of incomplete answers. You are not allowed to consult peers or use the web to aid in solving the problems. Columbia's academic honor code applies.

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Note: d, d_X, d_Y are used to denote generic metrics unless otherwise specified. If no metric is mentioned, assume the standard Euclidean metric d_2 .

1. **(40 points, (5 points each))** Are the following true or false? Briefly justify. If false, provide a counterexample when applicable.
 - (a) In a metric space (X, d) and for $A \subseteq X$, the set $A \setminus A'$ is open, where A' is the set of all limit points of A .
 - (b) In a metric space (X, d) and for $A \subseteq X$, if $x \in X$ is an interior point of A , then x is a limit point of A .
 - (c) In a complete metric space (X, d) and for closed $A \subseteq X$, then A is a complete set.
 - (d) In the metric space $(\mathbb{R}_{>0}, d_2)$, the set $(0, 1]$ is closed and bounded, and hence compact.
 - (e) Given metric spaces $(X, d_X), (Y, d_Y)$ and function $f : X \rightarrow Y$, if $f(K)$ is compact in (Y, d_Y) for any K compact in (X, d_X) , then f is a continuous function.
 - (f) In a real inner product space $(V, +, \cdot, \langle \cdot, \cdot \rangle)$, consider nonzero vectors $v_1, v_2 \in V$. If $\langle v_1, v_2 \rangle = 0$, then $v_1 - v_2, v_1 + v_2$ are linearly independent.
 - (g) Consider the vector space $(\mathbb{R}^2, +, \cdot)$ (under the usual componentwise definitions of vector addition and scalar multiplication). Then the function $\langle \cdot, \cdot \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$\langle x, y \rangle = x^T A y$$

is an inner product if $A \in \mathbb{R}^{2 \times 2}$ has all nonnegative entries.

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- (h) In a real normed vector space $(V, +, \cdot, \|\cdot\|)$ where V is a convex set, the norm is a function that is quasiconvex and homogeneous of degree 1.

Solution.

- (a) False. Consider metric space $(\mathbb{R}, d_2)$ and take A to be a singleton set. Then $A' = \emptyset$ and $A \setminus A' = A$, which is closed but not open.
- (b) False. Consider the discrete metric. Then every point is interior but no point is a limit point (pset 1 may be cited here).
- (c) True. Take any Cauchy sequence (x_n) in A . WTS $x_n \rightarrow x \in A$. Since (x_n) is in X , which is complete, $x_n \rightarrow x \in X$. Since A is closed, $x \in A$ (sequential characterization of closed).
- (d) False. Though closed and bounded in $(\mathbb{R}_{>0}, d_2)$, it is not compact. Couple ways to see this:
1. The open cover $\{(1/n, 2)\}_{n=2,3,4,5,\dots}$ has no finite subcover.
 2. In $(\mathbb{R}, d_2)$ the set is not closed and hence not compact, so it cannot be compact in the smaller space $(\mathbb{R}_{>0}, d_2)$.

- (e) False. Consider $(X, d_X) = (Y, d_Y) = (\mathbb{R}, d_2)$ and $f = \begin{cases} 0, & x \in \mathbb{R} \setminus \{0\} \\ 1, & x = 0 \end{cases}$. $f(K) \subseteq \{0, 1\}$ is a finite set and hence compact, but clearly f is not continuous (since $\forall \delta > 0, f(B_\delta(0)) = \{0, 1\} \not\subseteq (1/2, 3/2) = B_{1/2}(f(0))$).

- (f) True. Suppose

$$0 = \lambda_1 (v_1 - v_2) + \lambda_2 (v_1 + v_2).$$

Applying $\langle \cdot, v_1 \rangle$ to both sides yields

$$\implies 0 = (\lambda_1 + \lambda_2) \|v_1\|^2 \implies \lambda_1 + \lambda_2 = 0,$$

while applying $\langle \cdot, v_2 \rangle$ to both sides yields

$$\implies 0 = (\lambda_2 - \lambda_1) \|v_2\|^2 \implies \lambda_2 - \lambda_1 = 0.$$

Thus, $\lambda_1 = \lambda_2 = 0$.

- (g) False. Consider $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then $\langle x, x \rangle = x^T A x = 2x_1 x_2$ which is negative for $\text{sign} x_1 \neq \text{sign} x_2$, violating positive definiteness of an inner product.
- (h) True. $\forall x, y \in V, \lambda \in [0, 1], \mu \geq 0$ the norm is HOD 1 since

$$\|\mu x\| = \mu \|x\|,$$

and it is a convex function (and hence quasiconvex) by triangle inequality:

$$\|\lambda x + (1 - \lambda)y\| \leq \lambda \|x\| + (1 - \lambda) \|y\|.$$

2. **(8 points, (4 points each))** Consider nonempty set X having at least two elements, along with metrics on X denoted by $d : X^2 \rightarrow \mathbb{R}_{\geq 0}, d' : X^2 \rightarrow \mathbb{R}_{\geq 0}$. For all $x_1, x_2, y_1, y_2 \in X$, we define the functions

$$d_+ : X^2 \times X^2 \rightarrow \mathbb{R}_{\geq 0}, d_+((x_1, x_2), (y_1, y_2)) := \frac{1}{2}d(x_1, y_1) + \frac{1}{2}d'(x_2, y_2)$$

$$d_\times : X^2 \times X^2 \rightarrow \mathbb{R}_{\geq 0}, d_\times((x_1, x_2), (y_1, y_2)) := \sqrt{d(x_1, y_1) \cdot d'(x_2, y_2)}$$

(a) Is d_+ a metric?

(b) Is $d_\times$ a metric?

Solution.

Take any $x = (x_1, x_2), y = (y_1, y_2), z = (z_1, z_2)$ in X^2 .

(a) d_+ is a metric:

$$1. \quad 0 = d_+(x, y) \iff 0 = \frac{1}{2}d(x_1, y_1) + \frac{1}{2}d'(x_2, y_2) \iff d(x_1, y_1) = 0 \text{ and } d'(x_2, y_2) = 0 \iff x_1 = y_1 \text{ and } x_2 = y_2 \iff x = y.$$

$$2. \quad d_+(x, y) = \frac{1}{2}d(x_1, y_1) + \frac{1}{2}d'(x_2, y_2) = \frac{1}{2}d(y_1, x_1) + \frac{1}{2}d'(y_2, x_2) = d_+(y, x).$$

$$3. \quad d_+(x, y) + d_+(y, z) = \frac{1}{2}d(x_1, y_1) + \frac{1}{2}d'(x_2, y_2) + \frac{1}{2}d(y_1, z_1) + \frac{1}{2}d'(y_2, z_2) = \frac{1}{2}(d(x_1, y_1) + d(y_1, z_1)) + \frac{1}{2}(d'(x_2, y_2) + d'(y_2, z_2)) \geq \frac{1}{2}d(x_1, z_1) + \frac{1}{2}d'(x_2, z_2) = d_+(x, z).$$

(b) $d_\times$ is not a metric:

$$0 = d_\times(x, y) \iff 0 = \sqrt{d(x_1, y_1) \cdot d'(x_2, y_2)} \iff d(x_1, y_1) = 0 \text{ or } d'(x_2, y_2) = 0,$$

and choose $x_1 = y_1, x_2 \neq y_2$.

3. **(7 points)** Suppose function $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is given by

$$f(x) = \frac{1}{2} \log(1 + x).$$

(a) **(4 points)** Is f a contraction?

(b) **(3 points)** Given some $x_0 \in \mathbb{R}_{\geq 0}$, define sequence (x_n) recursively by

$$x_{n+1} = f(x_n).$$

Does $\lim_{n \rightarrow \infty} x_n$ exist? If so, compute it. If your answer depends on x_0 , please specify how it does so.

Solution.

(a) Yes. For any $x, y \in \mathbb{R}_{\geq 0}$, the mean value theorem implies there is some z between x and y such that

$$|f(x) - f(y)| = \left| \frac{1}{2} \log(1 + x) - \frac{1}{2} \log(1 + y) \right| = \left| \frac{1}{2} \frac{1}{1 + z} \right| |x - y| \leq \frac{1}{2} |x - y|.$$

(b) Yes. Note that $x_n = f^n(x_0)$ and by the contraction mapping theorem, since $(\mathbb{R}_{\geq 0}, d_2)$ is a complete metric space, $f^n(x_0)$ must converge to the unique fixed point of f for any $x_0 \in \mathbb{R}_{\geq 0}$, which is zero (i.e. $f(0) = 0$).

4. **(15 points)** Consider a family of matrices $\{A_t\}_{t \in \mathbb{R}_{\geq 0}}$, where

$$A_t = \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

- (a) **(8 points)** Is A_t diagonalizable in $\mathbb{C}$? Is A_t diagonalizable in $\mathbb{R}$? If so, diagonalize it, i.e. find a 2×2 invertible matrix P_t and diagonal matrix Λ_t s.t. $P_t^{-1}A_tP_t = \Lambda_t$. If your answer depends on $t \geq 0$, please be clear on how it does so.
- (b) **(3 points)** For what values of $t \geq 0$ is A_t positive semidefinite? Negative semidefinite? Indefinite?
- (c) **(4 points)** Define $t^* := \sup \{t : A_t \text{ is positive semidefinite}\}$. Compute $\lim_{n \rightarrow \infty} \frac{1}{2^{n-1}} A_{t^*}^n$.

Solution.

- (a) Since A_t is real and symmetric, the spectral theorem guarantees A_t is diagonalizable in $\mathbb{R}$ (and hence in $\mathbb{C}$). The characteristic polynomial is given by

$$P_{A_t}(\lambda) := \lambda^2 - \text{tr}(A_t)\lambda + \det(A_t) = \lambda^2 - 2\lambda + 1 - t^2 = (\lambda - (1 - t))(\lambda - (1 + t))$$

so the eigenvalues are

$$\lambda_{t,1} = 1 + t, \lambda_{t,2} = 1 - t.$$

The eigenvectors are characterized by

$$\ker(A_t - \lambda_{t,1}I) = \{k(1, 1) : k \in \mathbb{R}\}$$

$$\ker(A_t - \lambda_{t,2}I) = \{k(1, -1) : k \in \mathbb{R}\}$$

$$\implies P_t = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \Lambda_t = \begin{pmatrix} 1+t & 0 \\ 0 & 1-t \end{pmatrix}.$$

- (b) For $t \leq 1$, A_t is positive semidefinite. For $t > 1$, A_t is indefinite.
- (c) We have $t^* = 1$. You could compute $A_1^n = P_1 \Lambda_1^n P_1^{-1}$ or by induction it is easy to show $A_1^n = 2^{n-1} A_1$, $n \geq 1$. Thus, the limit in question is just A_1 .
5. **(15 points)** Consider the following function for positive integer n and real $\alpha > 0$:

$$F : \mathbb{R}_{>0}^{2n} \rightarrow \mathbb{R}, F(\mathbf{X}, \mathbf{Y}) = F(X_1, \dots, X_n, Y_1, \dots, Y_n) = \sum_{i=1}^n (X_i + Y_i)^\alpha.$$

- (a) **(4 points)** Is F homogeneous? If so, state its degree of homogeneity. Is F homothetic? What is

$$\left(\frac{1}{F(\mathbf{X}, \mathbf{Y})} \sum_{i=1}^n X_i \frac{\partial}{\partial X_i} F(\mathbf{X}, \mathbf{Y}) \right) + \left(\frac{1}{F(\mathbf{X}, \mathbf{Y})} \sum_{i=1}^n Y_i \frac{\partial}{\partial Y_i} F(\mathbf{X}, \mathbf{Y}) \right)?$$

- (b) **(3 points)** Is it true that for all $\alpha \in (0, 1]$, F is quasiconcave? Briefly justify.

- (c) **(5 points)** Log linearize the equation

$$Z = F(\mathbf{X}, \mathbf{Y})$$

around a point $(Z, \mathbf{X}, \mathbf{Y}) = (Z^*, \mathbf{X}^*, \mathbf{Y}^*)$ that satisfies the equation, where Z^*, X_i^*, Y_i^* are nonzero for all $i = 1, \dots, n$. That is, defining the log deviations¹ $\hat{Z} := \log(Z/Z^*)$, $\hat{X}_i := \log(X_i/X_i^*)$, $\hat{Y}_i := \log(Y_i/Y_i^*)$, $i = 1, \dots, n$, use a first order Taylor approximation for $Z - F(\mathbf{X}, \mathbf{Y})$ that is linear in $\hat{Z}, \hat{X}_i, \hat{Y}_i$ around the point where $\hat{Z}, \hat{X}_i, \hat{Y}_i$ are zero for all $i = 1, \dots, n$ and set this expression equal to zero to obtain an approximated equation. You may use shortcuts discussed in class.

- (d) **(3 points)** For the case $n = 1$, write a second order Taylor approximation for F around the point $(X_1, Y_1) = (a, b)$ for $a, b > 0$.

Solution.

- (a) Note that for $\lambda > 0$ we have

$$F(\lambda \mathbf{X}, \lambda \mathbf{Y}) = \sum_{i=1}^n (\lambda X_i + \lambda Y_i)^\alpha = \lambda^\alpha \sum_{i=1}^n (X_i + Y_i)^\alpha = \lambda^\alpha F(\mathbf{X}, \mathbf{Y})$$

so F is HOD α , and hence homothetic. Further, by Euler's theorem,

$$\left(\frac{1}{F(\mathbf{X}, \mathbf{Y})} \sum_{i=1}^n X_i \frac{\partial}{\partial X_i} F(\mathbf{X}, \mathbf{Y}) \right) + \left(\frac{1}{F(\mathbf{X}, \mathbf{Y})} \sum_{i=1}^n Y_i \frac{\partial}{\partial Y_i} F(\mathbf{X}, \mathbf{Y}) \right) = \alpha.$$

- (b) Since $(X_i, Y_i) \rightarrow X_i + Y_i$ is linear (hence, concave), and $t \rightarrow t^\alpha$ for $\alpha \in (0, 1]$ is an increasing concave transformation, then $(X_i, Y_i) \rightarrow (X_i + Y_i)^\alpha$ is concave. Finally, the sum of concave functions is concave so F is concave (hence quasiconcave).

- (c) We have

$$\begin{aligned} \hat{Z} &= \frac{1}{F(\mathbf{X}^*, \mathbf{Y}^*)} \sum_{i=1}^n (X_i^* + Y_i^*)^\alpha (\widehat{X_i + Y_i})^\alpha \\ &= \frac{1}{Z^*} \sum_{i=1}^n (X_i^* + Y_i^*)^\alpha \alpha (\widehat{X_i + Y_i}) \\ &= \frac{\alpha}{Z^*} \sum_{i=1}^n (X_i^* + Y_i^*)^\alpha \frac{X_i^* \hat{X}_i + Y_i^* \hat{Y}_i}{X_i^* + Y_i^*} \\ &= \frac{\alpha}{Z^*} \sum_{i=1}^n (X_i^* + Y_i^*)^{\alpha-1} (X_i^* \hat{X}_i + Y_i^* \hat{Y}_i) \end{aligned}$$

- (d) A second order approximation for $F(X_1, Y_1)$ is given by

$$\begin{aligned} &F(a, b) + F_{X_1}(a, b)(X_1 - a) + F_{Y_1}(a, b)(Y_1 - b) \\ &+ \frac{1}{2} \left(F_{X_1 X_1}(a, b)(X_1 - a)^2 + 2F_{X_1 Y_1}(a, b)(X_1 - a)(Y_1 - b) + F_{Y_1 Y_1}(a, b)(Y_1 - b)^2 \right) \\ &= (a + b)^\alpha + \alpha(a + b)^{\alpha-1}(X_1 + Y_1 - (a + b)) + \frac{1}{2}\alpha(\alpha - 1)(a + b)^{\alpha-2}(X_1 + Y_1 - (a + b))^2. \end{aligned}$$

¹Alternatively, you may define percent deviations $\hat{Z} := \frac{Z - Z^*}{Z^*}$, $\hat{X}_i := \frac{X_i - X_i^*}{X_i^*}$, $\hat{Y}_i := \frac{Y_i - Y_i^*}{Y_i^*}$, $i = 1, \dots, n$.

Alternatively, defining $W = X_1 + Y_1$ and $f : W \rightarrow W^\alpha$, use a single variable Taylor approximation about $a + b$ to obtain the same result:

$$f(a+b) + f'(a+b)(W - (a+b)) + \frac{1}{2}f''(a+b)(W - (a+b))^2.$$

6. **(15 points)** Consider the following problem:

$$\max_{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2} e^{\sqrt{x_1} + x_2}$$

$$\text{s.t. } x_1 + x_2 \leq m \quad (1)$$

where $m > 0$.

- (a) **(2 points)** Argue that a solution exists.
- (b) **(2 points)** Argue that constraint (1) must bind at any possible maximum.
- (c) **(8 points)** Solve the problem (i.e. obtain all maximizers (x_1, x_2)).
- (d) **(3 points)** The *value function* $v : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ is given by

$$v(m) := \max_{(x_1, x_2) \in \{(a, b) \in \mathbb{R}_{\geq 0}^2 : a+b \leq m\}} e^{\sqrt{x_1} + x_2}.$$

Compute its derivative function $v'(m)$ if it exists.

Solution.

- (a) The set $D(m) := \{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2 : x_1 + x_2 \leq m\}$ is nonempty, closed, and bounded in $(\mathbb{R}^2, d_2)$ and thus compact, and the objective function is a real-valued continuous function, so a solution exists by Weierstrass theorem (showing closed and bounded is not needed).
- (b) The constraint must bind since the objective function is strictly increasing in each argument (this in itself is an acceptable answer). More specifically, suppose not, and let $\epsilon := m - x_1^* - x_2^* > 0$ for maximizer (x_1^*, x_2^*) . Then $(x_1^* + \epsilon, x_2^*)$ is feasible and $f(x_1^* + \epsilon, x_2^*) > f(x_1^*, x_2^*)$, giving a contradiction.
- (c) To appeal to KKT, we rewrite the problem so that the universal set is open:

$$\max_{(x_1, x_2) \in \mathbb{R}^2} e^{\sqrt{x_1} + x_2}$$

$$\text{s.t. } g_1(x_1, x_2) := m - x_1 - x_2 = 0 \quad (1)$$

$$g_2(x_1, x_2) := x_1 \geq 0 \quad (2)$$

$$g_3(x_1, x_2) := x_2 \geq 0 \quad (3)$$

(Note: CQ in this context says $\nabla g_1(x_1, x_2) = (-1, -1), \nabla g_2(x_1, x_2) = (1, 0), \nabla g_3(x_1, x_2) = (0, 1)$ are linearly independent, which fails only when all constraints bind, which cannot happen since $m > 0$. Hence, CQ is always met).

Solution 1.

Set up the Lagrangian as

$$\mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, m) = e^{\sqrt{x_1} + x_2} + \lambda_1 (m - x_1 - x_2) + \lambda_2 x_1 + \lambda_3 x_2.$$

A maximizer must satisfy FOC and CSC:

FOC:

$$\partial_{x_1} \mathcal{L} = e^{\sqrt{x_1} + x_2} \frac{1}{2\sqrt{x_1}} - \lambda_1 + \lambda_2 = 0.$$

$$\partial_{x_2} \mathcal{L} = e^{\sqrt{x_1} + x_2} - \lambda_1 + \lambda_3 = 0.$$

CSC:

$$m - x_1 - x_2 = 0.$$

$$\lambda_2 x_1 = 0, \lambda_2 \geq 0, x_1 \geq 0.$$

$$\lambda_3 x_2 = 0, \lambda_3 \geq 0, x_2 \geq 0.$$

We have already argued (1) binds, and it is clear that (2) cannot bind as the FOC with respect to x_1 cannot be met then (acceptable to give economic interpretation of having infinite marginal utility). Thus, CSC implies $\lambda_2 = 0$ and there are only two cases:

Case 1. (3) binds. Then $x_2 = 0 \implies x_1 = m$ (by (1)) $\implies \lambda_1 = e^{\sqrt{m}} \frac{1}{2\sqrt{m}}$ (FOC x_1) $\implies \lambda_3 = e^{\sqrt{m}} \frac{1}{2\sqrt{m}} - e^{\sqrt{m}}$ (FOC x_2). This is a solution for $\lambda_3 \geq 0 \iff m \leq \frac{1}{4}$.

Case 2. (3) is slack. Then $x_2 > 0 \implies \lambda_3 = 0$ (by CSC) $\implies x_1 = 1/4$ (FOCs) $\implies x_2 = m - 1/4$ (by (1)) $\implies \lambda_1 = e^{m+1/4}$. This is a solution for $x_2 > 0 \iff m > \frac{1}{4}$.

Thus the unique maximizer is given by

$$x^* = (x_1^*, x_2^*) = \begin{cases} (m, 0), & m \leq \frac{1}{4} \\ (1/4, m - 1/4) & m > 1/4 \end{cases}.$$

While not necessary to state for this part, the multipliers are given by

$$\lambda^* = (\lambda_1^*, \lambda_2^*, \lambda_3^*) = \begin{cases} \left(e^{\sqrt{m}} \frac{1}{2\sqrt{m}}, 0, e^{\sqrt{m}} \left(\frac{1}{2\sqrt{m}} - 1 \right) \right), & m \leq \frac{1}{4} \\ (e^{m+1/4}, 0, 0) & m > 1/4 \end{cases}.$$

Solution 2.

Since the set of maximizers is unchanged by the strictly increasing transformation $t \rightarrow \log t$ applied to the objective function, one may replace the objective function with the function $(x_1, x_2) \rightarrow \sqrt{x_1} + x_2$, in which case the Lagrangian is given by

$$\tilde{\mathcal{L}}(x_1, x_2, \tilde{\lambda}_1, \tilde{\lambda}_2, \tilde{\lambda}_3, m) = \sqrt{x_1} + x_2 + \tilde{\lambda}_1 (m - x_1 - x_2) + \tilde{\lambda}_2 x_1 + \tilde{\lambda}_3 x_2.$$

A maximizer must satisfy FOC and CSC:

FOC:

$$\partial_{x_1} \tilde{\mathcal{L}} = \frac{1}{2\sqrt{x_1}} - \tilde{\lambda}_1 + \tilde{\lambda}_2 = 0.$$

$$\partial_{x_2} \tilde{\mathcal{L}} = 1 - \tilde{\lambda}_1 + \tilde{\lambda}_3 = 0.$$

CSC:

$$m - x_1 - x_2 = 0.$$

$$\tilde{\lambda}_2 x_1 = 0, \tilde{\lambda}_2 \geq 0, x_1 \geq 0.$$

$$\tilde{\lambda}_3 x_2 = 0, \tilde{\lambda}_3 \geq 0, x_2 \geq 0.$$

We have already argued (1) binds, and it is clear that (2) cannot bind as the FOC with respect to x_1 cannot be met then (acceptable to give economic interpretation of having infinite marginal utility). Thus, CSC implies $\tilde{\lambda}_2 = 0$ and there are only two cases:

Case 1. (3) binds. Then $x_2 = 0 \implies x_1 = m$ (by (1)) $\implies \tilde{\lambda}_1 = \frac{1}{2\sqrt{m}}$ (FOC x_1) $\implies \tilde{\lambda}_3 = \frac{1}{2\sqrt{m}} - 1$ (FOC x_2). This is a solution for $\tilde{\lambda}_3 \geq 0 \iff m \leq \frac{1}{4}$.

Case 2. (3) is slack. Then $x_2 > 0 \implies \tilde{\lambda}_3 = 0$ (by CSC) $\implies x_1 = 1/4$ (FOCs) $\implies x_2 = m - 1/4$ (by (1)) $\implies \tilde{\lambda}_1 = 1$. This is a solution for $x_2 > 0 \iff m > \frac{1}{4}$.

Thus the unique maximizer is given by

$$x^* = (x_1^*, x_2^*) = \begin{cases} (m, 0), & m \leq \frac{1}{4} \\ (1/4, m - 1/4) & m > 1/4 \end{cases}.$$

While not necessary to state for this part, the multipliers are given by

$$\tilde{\lambda}^* = (\tilde{\lambda}_1^*, \tilde{\lambda}_2^*, \tilde{\lambda}_3^*) = \begin{cases} \left(\frac{1}{2\sqrt{m}}, 0, \frac{1}{2\sqrt{m}} - 1 \right), & m \leq \frac{1}{4} \\ (1, 0, 0) & m > 1/4 \end{cases}.$$

(d) Few approaches:

Solution 3.

Direct computation of the indirect utility function gives

$$v(m) = e^{\sqrt{x_1^*} + x_2^*} = \begin{cases} e^{\sqrt{m}}, & m \leq \frac{1}{4} \\ e^{m+1/4} & m > 1/4 \end{cases}.$$

Thus,

$$v'(m) = \begin{cases} e^{\sqrt{m}} \frac{1}{2\sqrt{m}}, & m \leq \frac{1}{4} \\ e^{m+1/4} & m > 1/4 \end{cases}.$$

Solution 4.

Appeal to the envelope theorem (it is acceptable here to just perform the computation here without checking the conditions of the theorem). By solution 1 in the last part, labeling $x := (x_1, x_2, x_3), \lambda :=$

$(\lambda_1, \lambda_2, \lambda_3)$, the envelope theorem tells us

$$\begin{aligned} v'(m) &= \partial_m \mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, m)|_{(x, \lambda) = (x^*, \lambda^*)} \\ &= \lambda_1^* = \begin{cases} e^{\sqrt{m}} \frac{1}{2\sqrt{m}}, & m \leq \frac{1}{4} \\ e^{m+1/4} & m > 1/4 \end{cases}. \end{aligned}$$

Solution 5.

Appeal to the chain rule and envelope theorem (it is acceptable here to just perform the computation here without checking the conditions of the theorem). By solution 2 in the last part, labeling $x := (x_1, x_2, x_3)$, $\tilde{\lambda} := (\tilde{\lambda}_1, \tilde{\lambda}_2, \tilde{\lambda}_3)$, the envelope theorem tells us

$$\begin{aligned} v'(m) &= e^{\sqrt{x_1^* + x_2^*}} \partial_m \tilde{\mathcal{L}}(x_1, x_2, \tilde{\lambda}_1, \tilde{\lambda}_2, \tilde{\lambda}_3, m)|_{(x, \tilde{\lambda}) = (x^*, \tilde{\lambda}^*)} \\ &= e^{\sqrt{x_1^* + x_2^*}} \tilde{\lambda}_1^* = \begin{cases} e^{\sqrt{m}} \frac{1}{2\sqrt{m}}, & m \leq \frac{1}{4} \\ e^{m+1/4} & m > 1/4 \end{cases}. \end{aligned}$$